Omin Kwon

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Research Interest

Building efficient and scalable ML systems through algorithm-system co-design.

Education

Seoul National University Sept 2025 – present

M.S. in Computer Science and Engineering

- Advisor: Prof. Jae W. Lee, ARC Lab
- GPA: 4.3/4.3

Seoul National University Mar 2019 – Aug 2025

B.S. in Electrical and Computer Engineering (summa cum laude)

- GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)
- Ranked 1st in Introduction to ML, Computer Organization & Design, and Computer Networks.
- Leave of absence for service in the Republic of Korea Air Force (May 2021-Feb 2023)

Hansung Science High School Mar 2017 – Mar 2019

Early Graduation for Academic Excellence

Publications

SketchSSM: Write to the Full State, Read from a Compact Sketch
Omin Kwon, JoongWon Shin, Minseo Kim, Kurt Keutzer, Sehoon Kim, Jae W. Lee
NeurIPS 2026 Long-Context Foundation Model Workshop (Oral)
Under review at ICLR 2027 (Avg. Score: 3.0/4; 1,650/40,156 = 4.1%)

HERALD: High Throughput Block Diffusion LLM Inference via CPU-GPU Cooperative KV Cache Retrieval
Omin Kwon, Doyeon Kim, Jongseok Park, Seung Yul Lee, Sehoon Kim, Ion Stoica, Jae W. Lee
Under review at MLSys 2027

Triage: Skip, Skim, or Refine Attention Tiles in Video Diffusion Transformers
Yeonjae Kim, **Omin Kwon**, Seong Hoon Seo, Jae W. Lee
Under Review

MAGE: All-[MASK] Block Already Knows Where to Look in Block Diffusion LLM
Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee
NeurIPS 2026
ICML 2026 Adaptive Foundation Model Workshop (Oral, Best Paper Award (1/218 = 0.46%))

STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer
Omin Kwon, Kyungjun Oh, Jaeyong Lee, Myungsuk Kim, Jihong Kim
ICCAD 2025 (Oral)
U.S. patent application with SK hynix: No. 19/569899, filed March 17, 2026.

NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs
Haeun Lee, **Omin Kwon**, Yeonhong Park, Jae W. Lee
NeurIPS 2025

AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM
KyungSoo Kim, **Omin Kwon**, Yeonhong Park, Jae W. Lee
IEEE CAL

Research Experience

Graduate Student Researcher Sept 2025 – present
Seoul National University Seoul, South Korea

- Research visit to FAST Lab, UIUC (Mar-May 2026).
- **MAGE (Project Lead)** · Advisor: Prof. Jae W. Lee
  - Designed a training-free **sparse attention algorithm** for block diffusion LLMs.
  - Developed an **inference system** integrated into SGLang with optimized CUDA kernels.
  - Achieved up to 6.82× end-to-end speedup at 128K context with near-lossless accuracy.
- **HERALD (Project Lead)** · Advisor: Prof. Jae W. Lee
  - Designed a training-free **sparse attention algorithm** for block diffusion LLMs with KV offload.
  - Developed an **inference system** integrated into SGLang with a dual-stream CPU-GPU pipeline.
  - Achieved up to 2.28× decode throughput over GPU-only serving with near-lossless accuracy.
- **SketchSSM (Project Lead)** · Advisors: Prof. Jae W. Lee and Prof. Sehoon Kim
  - Designed a training-free **linear attention algorithm** that reduces state-read I/O.
  - Developed an **inference system** integrated into vLLM with optimized CUDA kernels.
  - Achieved up to 2.77× decode throughput over Standard vLLM with near-lossless accuracy.

Undergraduate Student Researcher Sept 2024 – Aug 2025
Seoul National University Seoul, South Korea

- **NestedFP** · Advisor: Prof. Jae W. Lee
  - Integrated the NestedFP CUDA kernel into vLLM with a workload-aware precision-switching scheduler.
- **AiDE** · Advisor: Prof. Jae W. Lee
  - Proposed workload-aware KV cache placement for CXL-PNM and developed an LLM simulator.
- **STAR (Project Lead)** · Advisor: Prof. Jihong Kim
  - Proposed a 3D NAND randomization algorithm that extends SSD lifetime by up to 2.3×.
  - Designed STAR hardware and integrated it into the SSD controller.
  - Key contributor to a U.S. patent application with SK hynix on STAR.

Teaching Experience

Teaching Assistant, Advanced Computer Architecture (4190.571) Sept 2026 – present
Seoul National University Seoul, South Korea

Undergraduate Course Tutor Mar 2024 – June 2025
Seoul National University Seoul, South Korea

- Selected as an official departmental tutor based on academic excellence.
- Courses: Computer Organization and Design, Digital System Design, and Microelectronics.

Honors & Awards

Best Paper Award @ ICML26 AdaptFM Workshop July 2026
ICML 2026 AdaptFM Workshop, Best Paper among 123 accepted papers from 218 submissions (0.46%).

Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest Jan 2025
Ministry of Science and ICT, designed a resolution-aware diffusion accelerator. News: Edaily.

Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest Dec 2024
Ministry of Education, designed a state-aware data randomizer for high-performance SSDs. News: Dong-A Ilbo.

Grand Prize, 2024 FriendliAI LLM Hackathon July 2024
FriendliAI, developed a knowledge-graph RAG application using the FriendliAI API.

SNU Specialized Semiconductor College Scholarship Aug 2024 – Aug 2025
Seoul National University

Merit-Based Scholarship 2023, 2024
Seoul National University